

Melbourne Airbnb Market Screening

Interim Project Report

CMCE30005 Business Analytics Challenge | Semester 2 2026 | TheNextChapter Group 2

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Introduction

Our client leases residential properties and runs them as Airbnb accommodation with the required permissions. This business model is called rental arbitrage. The client has limited start-up funds. They need to choose which Melbourne areas and property types to investigate before signing leases and buying furniture. A poor choice can tie up their money and delay the business. Our project helps the client narrow this search.

We use the Inside Airbnb dataset supplied by the school. It includes property details, locations and dated guest reviews. These fields allow us to compare existing listings. We group locations by Local Government Area (LGA), which means a municipal area. Each LGA can contain several suburbs. All numerical analysis uses the supplied dataset.

Our final output will be a shortlist of property segments for further investigation. We will compare their past review activity and test whether models improve on simple segment averages. The client can use the shortlist to guide property searches and request lease quotations. Financial assessment will require actual rent, furnishing costs and operating expenses, which this dataset does not contain.

Problem Definition and Objectives

Our research question is: Which LGA, dwelling-type and bedroom-count segments have the highest average predicted probability of reaching a common upper-quartile review benchmark among eligible Melbourne listings? Predictions are tested on hosts excluded from model training. The outcome covers the year before each listing's scrape date.

We study standard entire homes with one to three bedrooms. Rental units and condos form the apartment group; homes and townhouses form the house group. These broad residential groups match the client's search and reduce fragmented categories, although properties still differ within them. At least 50 eligible listings provides minimum comparison support. This is an analytical choice, so we also report independent host counts. A first review at least 365 days earlier limits short review histories without proving continuous operation. Quoted prices of AUD30–1,500 define an exploratory sample boundary. Sensitivity checks assess these restrictions; none establishes lease availability.

The target records whether a listing reaches a common review-count threshold over the previous 365 days. Dated reviews give us an observable measure of guest-review activity across the supplied listings. This supports an initial comparison of areas and property types. The upper quartile defines relatively high activity within our reference sample. Short stays can generate more reviews than long stays with similar guest nights. Review willingness also varies. These differences limit comparisons and prevent estimates of bookings, occupancy or profit.

We will deliver a segment comparison table, a model-versus-baseline assessment and a shortlist with evidence strength and uncertainty. These outputs guide property searches before lease commitment.

Data Description

The school dataset contains 25,728 listings with 90 variables, 9,390,720 calendar records and 1,026,690 reviews (Inside Airbnb, 2026). Scrape dates range from 17 June to 1 July 2026. We use LGA, property type, bedrooms, guest capacity, bathrooms, amenities and review dates. Calendar unavailability can include dates blocked by hosts. The calendar also contains no prices, so it cannot establish realised seasonal cash flow.

Cleaning converts price text into numbers, standardises missing values and extracts bathroom and amenity information. Price is missing for 6,553 listings and bedrooms for 4,679. We rebuild review counts over the preceding 365 days. GitHub records date-window checks and the sample-filtering table.

The separate reference sample contains 906 listings. Its 75th percentile is 29.75 reviews, giving an integer threshold of 30. The analysis sample contains 3,873 listings from 1,726 hosts in 14 segments. We keep 334 listings with zero annual reviews to include low-activity outcomes. Overall, 981 listings reach the threshold (25.33%).

Segment attainment ranges from 8.25% to 36.60%, supporting segment comparison. The sample covers six LGAs, with about 64% in Melbourne LGA. Holding the 30-review target fixed, removing the history rule gives 6,675 listings and 17.51% attainment. Removing the price restriction gives 5,720 listings and 18.29%. Both differ substantially from the primary 25.33%. Eligibility changes the population and its measured performance. These checks support limiting the shortlist to the stated sample; they have not established ranking stability under wider rules. Exited listings are absent.

Methodology and Analytical Approach

A fixed rule based on host identifiers reserves about 20% of hosts to set the benchmark. These reference hosts are kept separate from model training and evaluation. The threshold stays fixed across comparisons. We use five-fold validation and keep each host's properties in one fold. This prevents the same host appearing in training and validation (scikit-learn developers, n.d.). Each training fold learns its own missing-value treatment and category encoding.

Logistic regression provides a simple reference model. Forest and boosting models capture nonlinear relationships. Basic features include location, configuration, capacity, bathrooms and amenity counts. Six extensions add coordinates, beds and nine facilities. We exclude review-based predictors, ratings and host badges. Separate operating-condition models include current price and minimum stay, whose timing limits pre-opening interpretation.

We compare models with a simple segment-rate baseline. It uses each segment's attainment among training hosts, with a fixed smoothing weight of ten. AUC measures ranking ability. Average precision summarises identification of listings reaching the target. Brier score measures probability error; lower values are better. Calibration checks predicted probabilities against observed rates. R supports cleaning and tables. Python pipelines keep validation steps consistent across models.

We refit models under 20 further host splits to check split sensitivity. These runs reuse the same data. We assume current property details are relevant to preceding-year activity, although they may have changed during that year. Property-only models therefore remain historical comparisons. Current intervals hold predictions fixed and omit model-selection and benchmark uncertainty. Model choice remains exploratory, with no untouched final test.

Analysis Plan and Progress to Date

We replaced the original ROI and quarterly design with historical review analysis from the school snapshot. Weak basic-model results prompted the segment baseline. Cleaning, review reconstruction, descriptive analysis and grouped validation are complete. Table 1 uses the same sample and folds.

Table 1. Predictive performance

Predictor	AUC	Average precision	Brier score
Training-host segment rate	0.581	0.285	0.1868
Original property-only logistic	0.573	0.285	0.1886
Enhanced property-only forest	0.640	0.350	0.1813

Basic logistic and forest models underperform segment rates. The enhanced forest raises AUC by 0.059 and reduces Brier by 0.0055, a 2.9% reduction in probability error. Both improve in all five folds. We provisionally select it for the lowest property-only Brier; boosting has higher AUC at 0.647. Across 20 further host splits, forest AUC averages 0.626 (range 0.589–0.648), exceeding the segment baseline each time. The consistent, moderate gain supports further validation.

Table 2 answers the RQ using the enhanced forest's mean predictions. Each prediction comes from a model trained without that listing's host. All property-only models identify the same main top-three set as the segment baseline.

Table 2. Initial search candidates

Segment	Listings / hosts	Observed attainment	Mean predicted probability
Melbourne three-bedroom apartments	306 / 139	36.60%	34.45%
Melbourne two-bedroom apartments	1,235 / 482	32.79%	31.41%
Yarra Ranges three-bedroom houses/townhouses	90 / 77	30.00%	27.82%

Melbourne three-bedroom apartments rank first in all 20 further forest splits. The top-three set appears in 19. Yarra Ranges has a smaller evidence base. Split checks support the initial candidates, while wider-rule ranking sensitivity remains pending. Scores describe historical attainment among existing listings. Improvement within individual segments and future new-operator outcomes remain untested.

Segment rates provide a transparent starting shortlist. The forest adds property-level scores based on recorded attributes and improves overall prediction, with no change to the main candidates. We have not measured better investigation decisions or financial returns. The client can prioritise searches and lease enquiries in these segments, then check permissions, availability and actual costs before signing.

Next, inner training folds will select models and outer folds will evaluate that process. Refitted uncertainty estimates and wider dwelling/support rules will test the shortlist. We will report paired Brier differences, calibration-bin rates and candidate recurrence. Confirmed gains require lower Brier without worse calibration. Stable candidates across scope checks support stronger search priorities. If gains disappear, segment rates lead the screening. Unstable rankings produce a broader candidate list. These tasks remain planned; repeated splits have already tested one source of uncertainty.

References

AI assistance is acknowledged in the accompanying AI Use Declaration, also available in the repository.

Inside Airbnb. (2026). Melbourne listings, calendar and reviews [June 2026 dataset supplied through CMCE30005 LMS].

Scikit-learn developers. (n.d.). Cross-validation and probability calibration. Grouped validation; Calibration guidance